

3D-to-Technical Export

Build a standalone data-parser and rendering utility that converts a 3D scene into a multi-layered, professional-grade technical PDF or DXF.

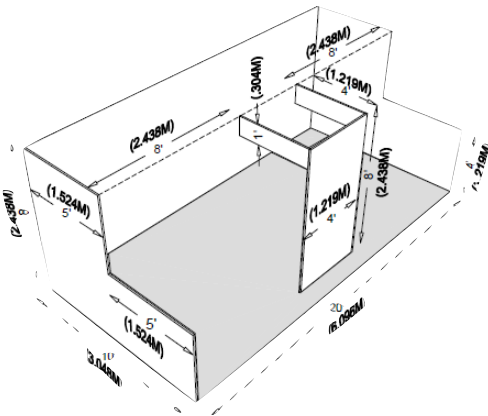
Using only as input a 3D scene containing

- Walls: Arrays of start/end coordinates, height, and thickness.
- Lighting/Rigging: Nodes for "Track Spotlights" and "Truss Segments" with specific height offsets.
- Furniture/Assets: Standard assets like tables, chairs, and "Lightbox" displays.

The Assignment Requirements

1) The Geometry & Dimensioning Engine

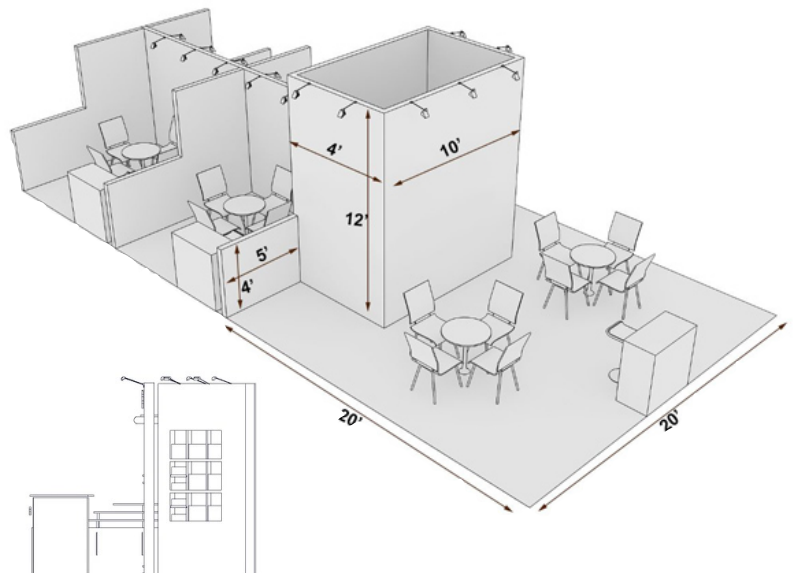
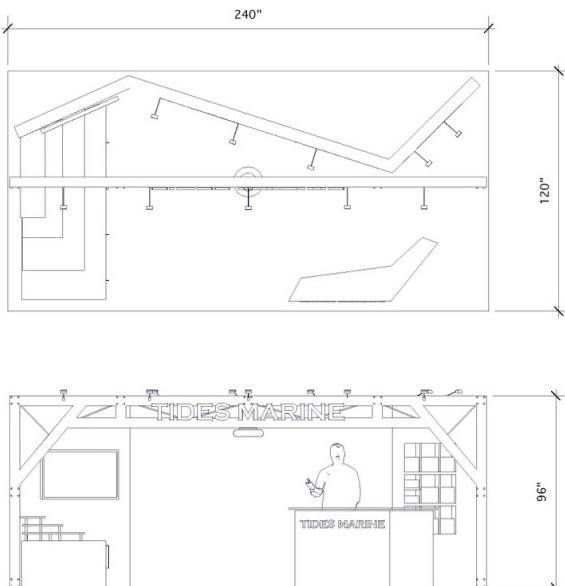
Implement the math to project the 3D scene onto a 2D (Top-down, isometric view, and Front-elevation views).



Write an algorithm to calculate and draw dimension lines (with arrowheads and text labels) for every wall and major asset, very much like the images below.

A preferred method is to project the dimensions directly onto the 3D asset and enable snapshots at different angles.

Ensure that dimension lines and text do not overlap with geometry or each other. Use standard drafting aesthetics: thin line weights, distinct tick marks/arrowheads for dimension lines, and clear, legible typography.



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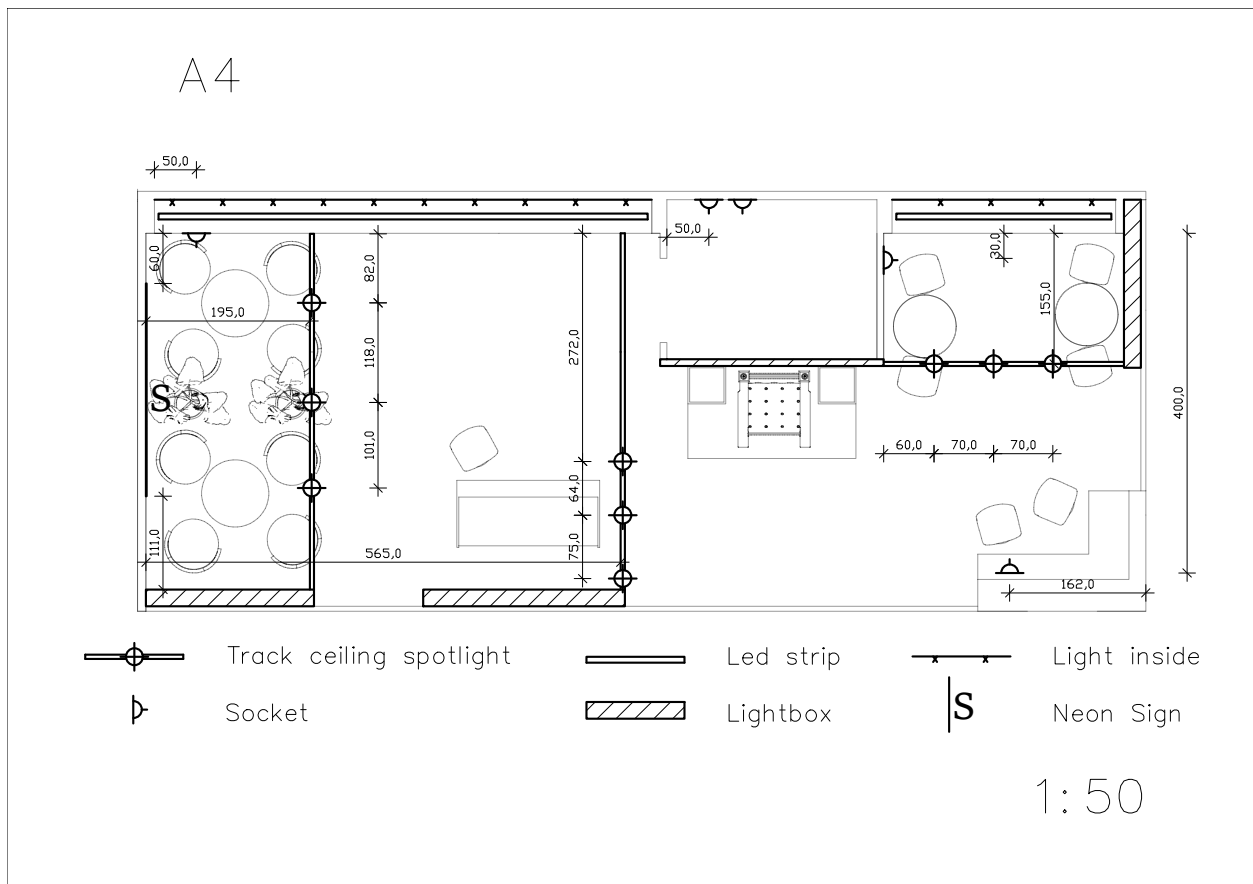
2) Layering, Legends & Professional Layout

The export must be generated with strictly separated layers that a user can toggle in a PDF/CAD viewer:

Layer 1: Architecture (Walls, floors, doors).

Layer 2: Lighting & Rigging (Truss, spotlights, LED strips).

Layer 3: Annotations (Dimensions, labels, and the legend).



How we measure your success:

Math Accuracy (Dimensions match, Handles unit conversions (Meters to Feet))
Visual Fidelity (Matches the "Drafting" aesthetic (line weights, arrowheads, title blocks))
Data structure (Layering - PDF/DXF has strictly managed, named layers)
Annotation (Basic text labels with intelligent labeling that avoids clutter and overlaps)

Note: this should work for pretty much any 3D asset that gets used as input, with a preference towards event context.